

SignSpeak

Features List

1. Sign Recognition (ASL → Text)

- Integrate the browser's camera API.
- Implement MediaPipe locally to extract hand landmarks directly in the browser
- Create an API endpoint to receive video frames from the frontend.
- Implement the ASL gesture recognition model to analyze the video frames.
- Create an API endpoint to send the translated text back to the frontend.

2. Conversation Interface

- Develop the UI component (the "synchronized panel") that displays the text translations received from the API.
- Design a simple, accessible layout with video and text panels.
- Implement Document PiP API

3. Future Possibilities

- **Speech Recognition (Voice → Text)**
 - Captures the other user's speech through the microphone.
 - Displays instant subtitles for deaf users.
- **Text-to-Speech**
 - Converts the signed text into spoken voice for hearing users
- **Low Latency AI**
 - Optimize the ML model to reduce inference time
- **Google Meet add-ons**
 - Integrate with Google Meet via Add-ons
- **Universal Picture-in-Picture (PiP) Interface**
 - Develop a Document Picture-in-Picture overlay that detaches the translation window.
 - Allow the window to float on top of any video conferencing platform (Meet, Zoom, Teams) without specific plugins.

MVP

Initial Hypothesis: Users will find value in a simple, accessible web app that translates American Sign Language (ASL) gestures into text, enabling smoother communication between deaf and hearing individuals during video calls.

MVP Features:

- Web interface with live webcam feed.
- ASL gesture recognition for basic signs and words.
- Display of translated text captions on screen.
- Lightweight backend handling video frame analysis via a ML model.

Not Included at MVP Stage:

- Complex sentence recognition or full ASL grammar understanding.
- Optional voice output using browser-based text-to-speech.
- Integration with Google Meet or other video platforms.
- User accounts, personalization, or conversation history.
- Advanced optimization for latency or multi-user sessions.

Goal: Validate whether users find real-time ASL-to-text translation useful and intuitive enough to assist in everyday communication in online meetings.

Outcome (Expected): Positive user feedback will confirm the potential of AI-powered sign language translation, leading to further iterations such as improved accuracy, smoother phrase recognition, and seamless integration with video conferencing platforms (e.g., Google Meet add-on).

MoSCoW Prioritization

Category	Features	Rationale
Must-Have	- Sign Recognition (ASL → Text) - Conversation Interface	These are the absolute minimum requirements for the core function: capturing ASL from a camera and displaying it as text.
Should-Have	- Integrate with Google meet add-on	These features provide significant value. Meet add-on integration brings the product directly into the user's workflow.
Could-Have	- Speech Recognition (Voice → Text) - Text-to-Speech - User accounts, personalization, or conversation history. - High Accuracy / Low	These are 'nice-to-have' improvements. They enhance the user experience (e.g., personalization, performance) but add complexity and are not required to test the core technology.

	Latency	
Won't-Have	- Support for other sign languages - Complex sentence recognition	These items are explicitly out of scope for this release. They represent different strategic goals (integration), significant technical complexity (AI improvement), or scope creep (new languages).

RICE Score

This table provides the RICE score for each feature. The final rank in the Product Backlog below is determined by a combination of this score and critical development dependencies.

Epic	MoSCoW	Specific Tasks Included	Reach	Impact	Confidence	Effort	Summed Effort	RICE Score
Testing and validating	Must	Unit, integration and system testing				10	10	
Conversation Interface	Must	Design layout and accessibility	1000	3	90%	10	60	45,0
		Develop UI component				30		
		Implement Document PiP API				20		
Integrate with Google meet add-on	Should	GCP Setup & Enable APIs	1000	2	60%	6	38	31,6
		Handle authentication				10		
		Create Manifest File				6		
		Develop Add-on manifest & UI				10		
		Handle deployment				6		
Text-to-Speech	Could	Integrate browser Text-to-Speech API	500	1	90%	10	15	30,0

		Implement voice API				5		
Sign Recognition (ASL → Text)	Must	Integrate camera API	1000	3	90%	14	140	19,3
		Build sliding buffer				10		
		Implement WebSocket Endpoint				6		
		Data Prep				20		
		Implement word classification model				70		
		Sentence Generator(LLM)				20		
Speech-to-Text (Voice → Text)	Could	Integrate browser Speech-to-Text API	500	2	80%	25	33	24,2
		Display captions in UI				8		
User Accounts & Personalization	Could	Develop backend (DB, Auth)	500	0,5	100%	15	25	10,0
		Develop UI (Login, Profile)				10		
High Accuracy / Low Latency	Could	Optimize the ML model	700	1	50%	50	50	7,0

Reordered Product Backlog for Release 1

PBI 1: Conversation Interface

- **RICE Score:** 45.0
- **User Story:** As a user (Rebecca), I want the translation interface to float over other applications (like Zoom or Meet), so I can see the text and the video call simultaneously without layout issues.
- **Reasoning:** This is a foundational **Must-Have** and has the highest RICE score. The application's core value proposition relies on being usable *during* video calls, making the PiP overlay critical for UX.

PBI 2: Sign Recognition (ASL → Text)

- **RICE Score:** 19.3
- **User Story:** As a deaf student (Rebecca), I want my signs translated into text in a simple web app during my Google Meet class for others to read.
- **Reasoning:** This is the core **Must-Have** engine of the product. It is a **critical dependency**. Its RICE score is low only because of its very high Effort (122 hours), but it must be developed in parallel with PBI 1.

PBI 3: Testing and Validating

- **RICE Score:** (N/A - Mandatory Process)
- **User Story:** As a Product Owner, I want to perform unit, integration, and system testing, so that I can ensure the final release is stable, bug-free, and meets all acceptance criteria.
- **Reasoning:** This is a **Must-Have** process. It is the final phase required to validate the other PBIs before the project deadline.

PBI 4: Text-to-Speech (Voice Output)

- **RICE Score:** 30.0
- **User Story:** As a deaf student (Rebecca), I want my translated text to be converted to speech, so that my hearing classmates can understand me easily.
- **Reasoning:** This "Could-Have" is included because it has the next highest RICE score and the low effort makes it a feasible.

Gantt Chart

